

Miguel L. Rodrigues

Computer Vision & Control Systems Engineer

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Mechatronics engineer between **industrial computer vision** and **control theory**: production deep-learning inspection pipelines owned end to end, and research on **event generators for saturated LPV systems** — LMI synthesis, Lyapunov certificates. Published at ROCOND, SBAI, and CBA.

■ EXPERIENCE

AI / Computer Vision Engineer Oct 2024 – Present
Digimet Solutions — Joinville, SC (remote)

- **Weld-bead inspection** in production — segmentation and keypoint models feeding geometric measurement.
- End-to-end ownership: annotation infrastructure (Label Studio · EC2 · PostgreSQL · S3), training, validation, mask post-processing.
- Serverless inference on **AWS Lambda** with CloudWatch; ONNX and INT8 quantization for latency and cost; domain adaptation for field imagery.

Undergraduate Researcher (scholarship & volunteer) Feb 2021 – Mar 2024
CEFET-MG — Departments of Mechatronics & Computer Engineering

- Controller synthesis and Lyapunov analysis for linear systems; Takagi–Sugeno neuro-fuzzy modeling; AI-based control of manipulators (KUKA youBot).
- Fuzzy inference for investment recommendation, ML for virtual drug screening, imaging for prostate-brachytherapy automation.

Teaching Assistant Mar – Sep 2024
CEFET-MG — Signals & Systems; Control Theory

■ RESEARCH & PUBLICATIONS

	Event generators that cut transmissions in networked loops while certifying stability — LPV models, saturation, LMIs, evolutionary co-design.	
EVENT-TRIGGERED & ROBUST CONTROL	Co-Design and Evolutionary Optimization of Periodic Event-Triggered Robust Control for LPV Systems. Oliveira, Rodrigues , Guelton, Motchon, Leite.	ROCOND 2025
	Gerador Dinâmico de Eventos com Tratamento de Ruído para Redução de Transmissões. Rodrigues , Silva, Leite.	SBAI 2025
	Gerador de Eventos para Minimização de Transmissões em Sistemas LPV Saturantes. Rodrigues , Silva, Leite.	CBA 2024
	Modelagens e Simulações Computacionais Aplicadas ao Desenvolvimento Remoto de Sistemas Mecatrônicos. Silva, Alves, Rodrigues , Dâmaso.	CBA 2022
MECHATRONICS & SIMULATION		

■ SELECTED PROJECTS

	robotic_tools — serial manipulators from Denavit–Hartenberg parameters: forward/inverse kinematics, Euler–Lagrange dynamics, Jacobians, gradient-based and evolutionary IK.
ROBOTICS & MECHANICS	beam_tools — symbolic Euler–Bernoulli beams via Macaulay singularity functions: closed-form shear, moment, slope, deflection; determinate and hyperstatic.
	prometheus — ESP32 temperature loop with UFMG’s GREa group: discrete PID with anti-windup, 2nd-order IIR pre-filter, open-to closed-loop handover, dual-core FreeRTOS, MQTT telemetry.
EMBEDDED CONTROL	fpwm-signal-generator — ESP32 function generator (ESP-IDF/C): sine, sawtooth, triangle, square via PWM and hardware timers; under 1% frequency error.

■ EDUCATION

M.Sc., Electrical Engineering (in progress)	2025 – 2027
CEFET-MG — Event-triggered control of LPV systems with actuator saturation	
B.Eng., Mechatronics Engineering	2020 – 2025
CEFET-MG — Control systems, automation, robotics, embedded systems	

■ SKILLS & LANGUAGES

VISION & AI	PyTorch · OpenCV · semantic & instance segmentation · keypoint detection · ONNX · INT8 quantization · domain adaptation
CONTROL	LPV systems · event-triggered & robust control · LMIs · Lyapunov stability · NumPy · SciPy · SymPy
EMBEDDED	ESP32 · STM32 · ESP-IDF · FreeRTOS · KiCad · power electronics
CLOUD & CODE	AWS (Lambda, EC2, S3) · Docker · CloudWatch · CI · Python · C · TypeScript
LANGUAGES	Portuguese (native) · English (full professional) · French (elementary)